

10-Day Learning Roadmap: AWS, PostgreSQL, Redis, API Structuring, Golang

Day 1: AWS Basics - Part 1

- Understand what AWS is (IaaS, PaaS, SaaS)
- Explore AWS global infrastructure (Regions, AZs, Edge locations)
- Set up an AWS Free Tier account
- Get familiar with IAM (users, roles, policies)
- Launch your first EC2 instance

Day 2: AWS Basics - Part 2

- Learn about S3: buckets, objects, permissions
- Understand VPC basics: subnets, gateways, route tables
- Intro to RDS and deploying PostgreSQL on RDS
- Monitor with CloudWatch and set basic alarms
- Basic billing dashboard understanding

Day 3: PostgreSQL - Part 1

- Install PostgreSQL locally or access via RDS
- Learn basic SQL: SELECT, INSERT, UPDATE, DELETE
- Understand schema, tables, data types
- Use pgAdmin or DBeaver for GUI access
- Practice creating and querying sample databases

Day 4: PostgreSQL - Part 2

- Learn constraints: PRIMARY KEY, FOREIGN KEY, UNIQUE, CHECK
- Understand indexes and their importance
- Learn JOINS: INNER, LEFT, RIGHT, FULL
- Basic performance tuning tips
- Use transactions and understand ACID properties

Day 5: Redis Basics

- What is Redis and why use it (caching, in-memory DB)
- Install Redis locally
- Learn basic commands: SET, GET, DEL, EXISTS, EXPIRE

- Understand Redis data types: strings, lists, sets, hashes
- Use Redis for caching a database query

Day 6: API Structuring - Part 1

- Understand RESTful API design principles
- Resource naming conventions (nouns, plural)
- Use of HTTP methods: GET, POST, PUT, DELETE
- Intro to status codes and response formats
- Set up basic folder structure for an API project

Day 7: API Structuring - Part 2

- Understand versioning (/api/v1)
- Middleware concepts (logging, error handling)
- Request validation and sanitization
- Authentication basics (token-based, sessions)
- Pagination, filtering and sorting in APIs

Day 8: Golang - Part 1

- Install Go and set up workspace (GOPATH, modules)
- Understand package structure
- Basic syntax: variables, functions, conditionals, loops
- Data types and control structures
- Write your first CLI Go app

Day 9: Golang - Part 2

- Structs and interfaces
- Error handling idioms
- Working with slices, maps, pointers
- Concurrency with goroutines and channels
- Practice building small modules

Day 10: Integration & Review

- Create a mini API using Go + PostgreSQL + Redis
- Deploy on AWS EC2 or ECS

- Use S3 to store static assets (optional)
- Practice explaining your project design
- Review notes and solidify weak areas